Major Project Stage-1 First Review Presentation
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Predictive Thermal Battery Management System for Li-ion Battery Packs

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Contents

1. Abstract
2. Problem Statement
3. Literature Survey
4. Problem Analysis / Research Gap
5. Objectives
6. Methodology
7. Block Diagram / Schematic Diagram
8. Hardware and Software Requirements
9. Project Timeline
10. References

Abstract

- Lithium-ion cells store a lot of energy in a small space. A cell that overheats can go into thermal runaway, where the heat it makes drives it hotter still.

- Most low-cost battery protection boards act only on voltage and current, and treat temperature, if at all, as one fixed limit that is reached late.

- This project builds a predictive thermal battery management system for a 4S1P pack of NCR18650GA cells, with two protection layers that work independently.

- A hardware BMS gives over-voltage, under-voltage, over-current and short-circuit protection. An ESP32 measures every cell voltage, the pack current and the cell and air temperatures, and uses a thermal model to predict how fast each cell should heat.

- When a cell heats faster than the current explains, or a reading crosses a limit, the ESP32 opens the charge or load relay and latches the fault. Readings are shown on a Wi-Fi dashboard that is not part of the safety chain.

Problem Statement

- Lithium-ion packs are used in more and more devices, and one overheating cell can lead to fire or explosion through thermal runaway.
- Most low-cost protection boards act only on voltage and current. Temperature, if it is checked at all, is compared with one fixed limit.
- A fixed limit acts only once the cell is already hot. A cell that is steadily heating at a normal temperature passes it with no warning.
- When protection depends on a single microcontroller, a firmware crash or a failed sensor can leave the pack unprotected.
- A system is needed that watches the temperature trend, stops only the unsafe direction of current, and still protects the pack if the software fails.

Literature Survey

S.No	Paper Title	Outcomes	Limitation / Research Gap
[1]	Thermal runaway mechanism of lithium ion battery for electric vehicles: A review	Reviewed how thermal runaway starts and spreads through a chain of heat-producing reactions inside the cell.	Explains the mechanism, but does not give an on-board method to detect it early.
[2]	Battery management system: An overview of its application in the smart grid and electric vehicles	Showed that a BMS needs accurate state estimation and ways to protect the battery from hazardous conditions.	Focuses on state-of-charge and state-of-health estimation, not on early warning from temperature.
[3]	Protection devices in commercial 18650 lithium-ion batteries	Opened four commercial 18650 cells. Only the current interrupt device and the top vent are fitted in every cell.	These are last-resort devices inside the cell. Pack-level monitoring and early warning are not covered.
[4]	Advanced fault diagnosis for lithium-ion battery systems: A review of fault mechanisms, fault features, and diagnosis procedures	Reviewed internal battery faults, sensor faults and actuator faults, and how each one can be diagnosed.	A review only. It does not build or test a low-cost diagnostic system.
[5]	A critical review of thermal runaway prediction and early-warning methods for lithium-ion batteries	Grouped early-warning methods into electrochemistry-based, big-data and artificial intelligence methods.	The groups it describes rely on detailed models, large data sets or AI training, not a simple on-board check.

Literature Survey (contd.)

S.No	Paper Title	Outcomes	Limitation / Research Gap
[6]	An online data-driven fault diagnosis and thermal runaway early warning for electric vehicle batteries	Found the cell heading for thermal runaway before it happened, using voltage and temperature data from real vehicles.	Compares many cells statistically, which suits large EV packs rather than a four-cell pack.
[7]	A neural network based method for thermal fault detection in lithium-ion batteries	A neural network predicts the cell surface temperature. A large gap between prediction and reading flags a fault.	Needs neural network training and retraining, which is heavy for a low-cost microcontroller.
[8]	A combined data-driven and model-based algorithm for accurate battery thermal runaway warning	Combined K-Means clustering with the Bernardi heat equation and warned 25 minutes before thermal runaway.	Needs training data and fitted parameters. Its 1 °C/s limit confirms runaway; it is not an early warning.
[9]	Online parameterization of lumped thermal dynamics in cylindrical lithium ion batteries for core temperature estimation and health monitoring	Showed that only the surface temperature can be measured, while the core can be hotter, and estimated the core.	Needs online parameter identification, and was tested on one 26650 LFP cell, not an 18650 pack.
[10]	Cloud-based battery condition monitoring platform for large-scale lithium-ion battery energy storage systems using internet-of-things (IoT)	Sent battery module data over IoT links to Google Cloud, where the cell states were worked out.	Depends on the cloud and a network link. Protection has to keep working when Wi-Fi is lost.

Problem Analysis

- Existing studies explain how thermal runaway builds up through heat-producing reactions [1], and show the cell core can be hotter than its surface [9].
- Early-warning methods mostly rely on detailed models, large data sets, neural networks or the cloud [5]–[8], [10].
- The protection board used here acts on voltage and current only. It does not watch temperature at all.

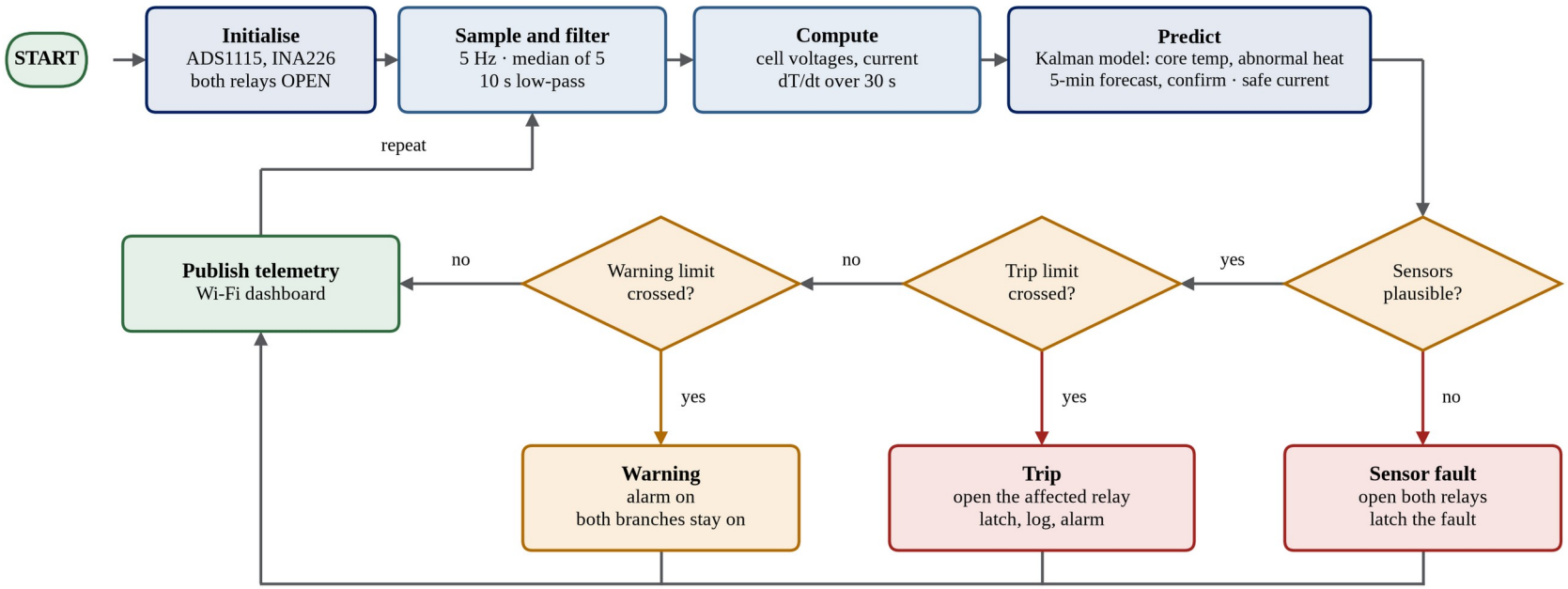
Research Gap

- A simple method is needed that spots an abnormal temperature rise early, using only cell-surface sensors and a low-cost microcontroller.
- The early warning must be backed by protection that still works if the controller, a sensor or the network fails [4].
- Charging and discharging should be cut off separately, so that only the unsafe direction is stopped.

Objectives

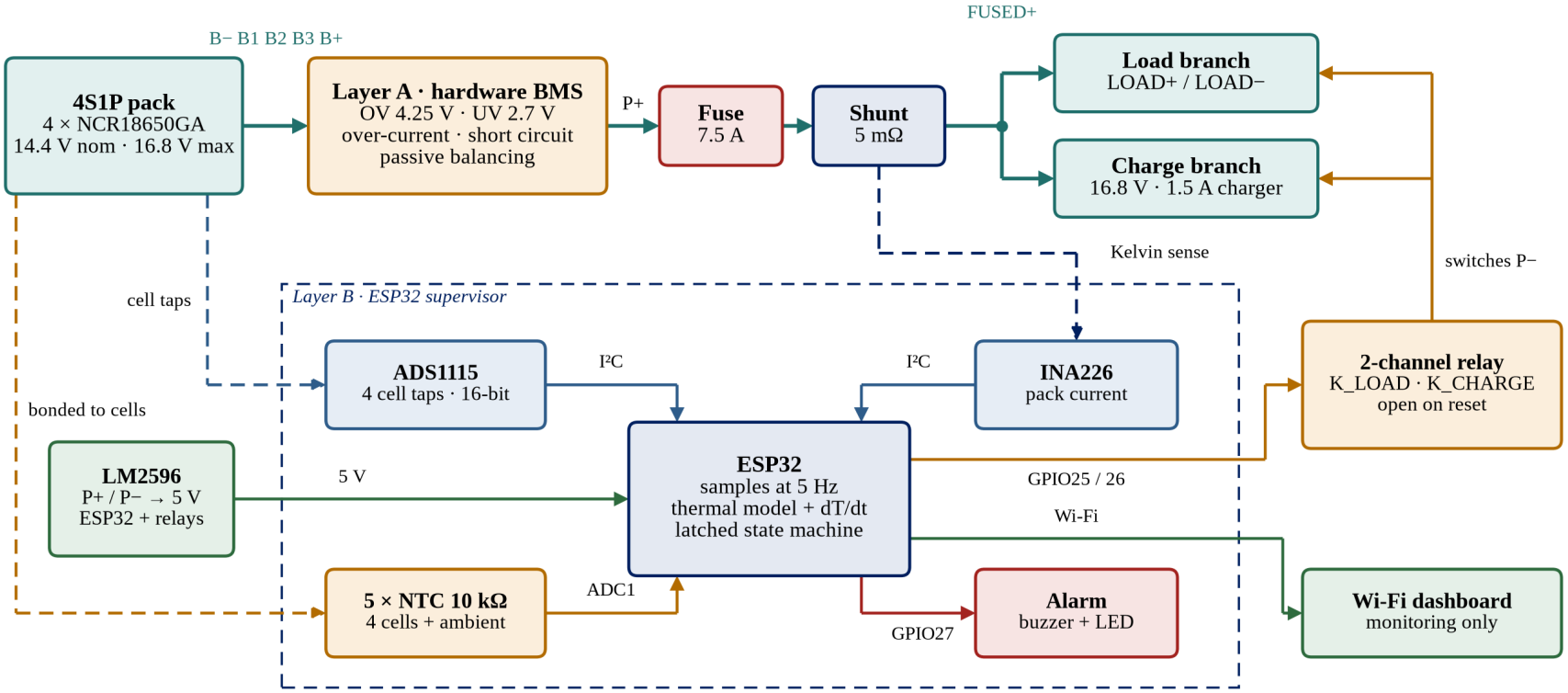
- To study thermal runaway in lithium-ion cells and the methods used to detect it early.
- To design a two-layer protection system: an independent hardware BMS, and an ESP32 supervisor that can disconnect charging and discharging separately.
- To measure every cell voltage, the pack current, four cell temperatures and the air temperature accurately, using an ADS1115, an INA226 and NTC thermistors.
- To predict each cell's heating with a thermal model, and act on heating the current cannot explain before any fixed temperature limit is reached, with every trip latched.
- To show live readings and faults on a Wi-Fi dashboard, while every safety function keeps working without Wi-Fi.

Methodology



Every trip stays latched. Manual reset only after 60 s healthy: all cells 3.20–4.10 V and all NTCs below 40 °C.

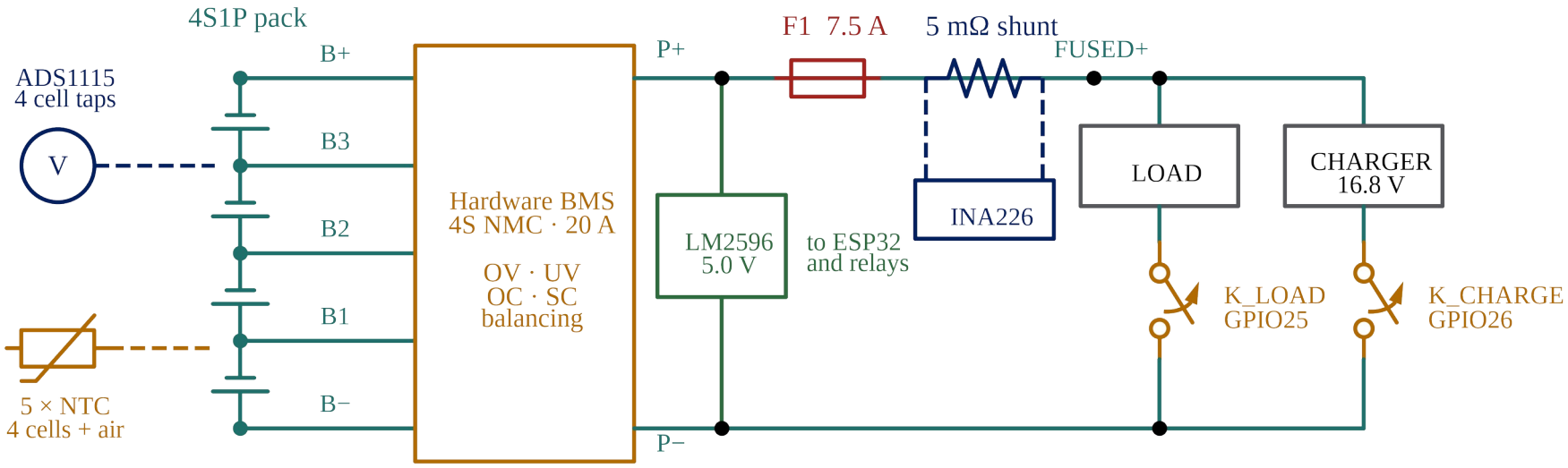
Block Diagram



Layer A protects the cells on its own. Layer B adds early warning and separate charge and load control, and opens both relays if it fails.

Schematic Diagram

- The hardware BMS protects the cells on its own, even if the ESP32 or Wi-Fi fails.
- Each relay is normally open and held off by a pull-up, so a reset disconnects both branches.



Hardware and Software Requirements

Type	Item	Used for
Hardware	ESP32-WROOM-32 DevKit	Sensing, thermal prediction, relays, Wi-Fi
Hardware	4 × NCR18650GA 3300 mAh	The 4S1P pack, with a 16.8 V 1.5 A charger
Hardware	4S 20 A hardware BMS	Voltage, current and short-circuit protection
Hardware	ADS1115 16-bit ADC	The four cell-tap voltages
Hardware	INA226 with 5 mΩ shunt	Pack current, charging and discharging
Hardware	5 × NTC 10 kΩ B3950	Temperature of each cell and of the air
Hardware	2-channel 5 V relay module	Separate charge and load disconnect
Hardware	LM2596, 7.5 A fuse, buzzer	5 V supply, wiring protection, local alarm
Software	ESP-IDF with C++	Firmware framework and language
Software	FreeRTOS	Sensor, safety and telemetry tasks
Software	HTML, CSS, JavaScript	The Wi-Fi dashboard
Software	FreeCAD, Git and GitHub	Enclosure model, version control

Project Timeline

Activity	Month 1	Month 2	Month 3	Month 4	Month 5
Objective 1: Literature Survey & System Study	√	√			
Objective 2: Two-Layer Design & Component Selection	✗	√			
Objective 3: Sensing Hardware, Pack Assembly & Calibration	✗	✗			
Objective 4: Thermal Prediction & Safety State Machine	✗	✗			
Objective 5: Wi-Fi Dashboard, Testing & Documentation	✗	✗			

√ = Completed
✗ = Yet to be done

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Thank You

Questions are welcome